Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 1st Term Examination, 2020.

URP 1113

(Fundamentals of Planning Process)

Full Marks: 120 Time: 1.5 hrs

N.B. i) Answer any **two** questions from each section in separate script.

ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Briefly discuss about the following: (20)
(i) Top-down approach, (ii) Inclusive city, (iii) Spatial Planning, (iv) Elements of a comprehensive plan.

(b) Define planning. Discuss the historical origins and evolution of Urban Planning in short. (10)

2. (a) Suppose, you have to implement a land use plan in a spatial planning zone of a city and have to consult with the stakeholders accordingly. How do you deal with the issue from the planning perspectives? (20)

(b) Compare between “Rational Planning” and “Advocacy Planning” with relevant example. (10)

3. (a) Give a concise review of “Funnel Approach” of Planning theory with a practical example. (15)

(b) How does the sustainability prevail in Planning Practice? Draw the sustainability Pillar to execute the process. (15)

Section – B

4. (a) What is urban sprawl? Discuss the end results of urban sprawl in a metropolitan city with necessary examples. (10)

(b) What is a primate city? Write down the advantages and disadvantages of a primate city. (10)

(c) “Urban planning has three dimensions in terms of Space, Sector and Time”. Briefly explain with relevant examples in context of Bangladesh. (10)

5. (a) Briefly explain the terms Vision, Mission, Goal and Objective with relevant examples especially in context of Urban and Regional Planning. (15)

(b) What is development plan? Explain four stages plan for a city (a city is considered at a local scale) with necessary sketches. (15)

6. (a) Why is your city different from other cities in terms of layout, size and scenic beauty? Briefly explain with necessary examples. (10)

(b) Suppose, you are working as a planner in Khulna WASA. Khulna WASA needs to develop a drainage system for Khulna City. Discuss the stages of conventional planning process for the above mentioned scenario. (20)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 1st Term Examination, 2020

Ch 1117

(Environmental Chemistry)

Full Marks: 120 | Time: 1.5 hrs

N.B. i) Answer any **two** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Why Tertiary Carbocations are more stable than Primary Carbocations? (08)
   (b) An S_N2 reaction proceeds with complete stereochemical inversion - Explain. (10)
   (c) Discuss the stereochemistry of S_N1 reaction. (12)

2. (a) What is living polymers? Give examples. (08)
   (b) Whether polyacetylene is a conducting polymer or not? Explain with suitable structure. (10)
   (c) Briefly discuss the structure and properties of isotactic, syndiotactic and atactic polypropylene. (12)

3. (a) Graphically represent the major regions of atmosphere with temperature profile and write the important chemical species in each step. (10)
   (b) Discuss the biochemical effect of SO_X and NO_X. (08)
   (c) What is meant by corrosion? Discuss the prevention of corrosion by modification of environment. (12)

Section – B

4. (a) Let a complex compound $[Co(en)_2(Cl)_2]NO_3$ (10)
   (i) Find out the coordination and oxidation numbers of the central metal atop;
   (ii) Find out the bidentate ligand; and
   (iii) The effective atomic number (EAN) of the central metal atom.
   (b) Compare the magnetic properties of $[Fe(CN)_6]^{4-}$ and $[Fe(H_2O)_6]^{2+}$ complexes with the help of Valence bond theory. (12)
   (c) Transition metals generally form colour compounds - why? (08)

5. (a) What do you mean by binding energy of nucleus? Show the relationship between binding energy and nuclear stability. (12)
   (b) Compare the following nuclear reactions (08)
   (i) $^{212}_{84}Po \rightarrow {}^{208}_{82}Pb + ?$
   (ii) $^{137}_{55}Cs \rightarrow {}^{137}_{56}Ba + ?$
   (c) Write short note on nuclear hazards. (10)

6. (a) What is bond order? Write down the conditions for effective combination of atomic orbitals to form a molecule. (10)
   (b) How does intermolecular and intramolecular hydrogen bonding influence the physical and chemical properties of the compounds? (10)
   (c) The energy of the 2p subshell in F in lower than the energy of 1s orbital in H. Why? (10)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 1st Term Regular Examination, 2020

HUM 1117

(Functional English)

Full Marks: 120 Time: 1.5hrs

N.B. i) Answer any TWO questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Make sentences on the following structures using the verb given in brackets. (10)
   - (i) Subject + Intransitive verb + Adverbial of place + Adverbial of time (work as verb)
   - (ii) Subject + Linking verb + Noun complement + Preposition phrase (remain as verb)
   - (iii) Subject +Transitive verb + Gerund as object (stop as verb)
   - (iv) Subject + Transitive verb + Object + Object (send as verb)
   - (v) There + Subject + Extension (exist as verb)

   (b) Change the following words asked in brackets and make sentence with the changed forms. (10)
   Efficiency (into adj.), Consistency (into adj.), Beneficence (into adj.), Impression (into verb), Dissatisfaction (into verb)

   (c) Make use of the following modals in sentence as asked in brackets. (10)
   - (i) Can. (To express a polite request).
   - (ii) May. (To give someone a permission).
   - (iii) Could. (To express a permanent ability in the past).
   - (iv) Shall. (To propose someone something).
   - (v) Must + have + past participle of verb. (To express a logical deduction)

2. (a) Transform the following sentences as asked in brackets. (10)
   - (i) Nasim, a banker, works hard. (complex).
   - (ii) He can progress because of his perseverance. (complex).
   - (iii) Mim is as hard working as Shimu. (comparative).
   - (iv) When Liza was twelve years old, she could run much. (simple).
   - (v) Simu studies hard, but can't do well in exam. (simple).

   (b) Make sentences using the following words as directed. (10)
   what(as adjective); what (as pronoun); while (as noun);while (as verb); while (as adverb).

   (c) Complete the sentences with subordinate clauses. (10)
   - (i) He seemed ………………………………….……………..… (noun clause)
   - (ii) He walked ten miles…………………………….….…. (adjective clause)
   - (iii) …………………………………he disappeared. (adverb clause of time)
   - (iv) The committee was formed……………….. (adverb clause of purpose)

3. (a) Frame wh questions from the underlined parts of the following answers. (10)
   - (i) Nobody believes <u>a liar.</u>
   - (ii) They will go to Dhaka <u>by air.</u>

(iii) I like to play <u>cricket.</u>

(iv) <u>Smuggling goods</u> is a heinous crime.

(v) The door was opened <u>by a servant.</u>

(b) Correct the following sentences. (10)

(i) Chairman is supposed to preside over the meeting.

(ii) The football is a popular game.

(iii) My pen is better than your.

(iv) There all fishes are rotten.

(v) He is better than me.

(c) Make sentences expressing the following notions/emotions. (10)

(i) Consolation (ii) Disapproval (iii) Possibility (iv) Sympathy (v) Introduction

Section – B

4. (a) Read the following passage carefully and answer the questions that follow. (18)

Natural calamity affects severely the life of human being, their life goes deadlock. Hopelessly they suffer, feel unimagined miseries, become socially separated, lead life ego centrically and so play the role of existential philosophy. For Khawaja the existential philosophy is "the idea of personal authenticity is at the centre of essential thought." So it emphasizes individual existence, freedom and choice. Nobody is for others in a society. Whatever a man suffers can't create any effect in any one neighbor, but it can't be and is the life of a human being. Man is for man, they share their happiness, miseries, love, etc. in any situation. But national calamity often results from the activity of human beings. we without a little consideration of our conscience cut threes, build habitation, contaminate water, air and soil, make industries, though we sometimes or many times do these for our necessity, they prepare the way of imbalance in everysphere of natural environment. Resultantly natural hazard is born in the earth. Let's be aware of our life as is in the idea of T. S. Eliot, a modern English Poet, the life of a human being is on the verge of destruction because of the pollution of environment 'The river sweats/oiled and tar.' (The Waste Land).

Questions:

1. How is life in natural environment?
2. What is existential philosophy?
3. How is a man responsible for natural hazard?

(b) Make a precis of the above written passage with a suitable title. (12)

5. (a) Write a paragraph on a great soul of a human being. (18)

(b) Amplify the idea contained in the following statement. (12)

Time is money.

6. (a) i. Write a report on the medical centre of your varsity. (18)

ii. Prepare a C.V. along with a job application. (12)

or

(b) Write a free composition on the importance of a planned city and the role of planners. (30)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 1st Term Examination, 2020

Math 1117

(Mathematics-I)

Full Marks: 120 Time: 1.5hrs

N.B. i) Figures in the right margin indicate full marks.

ii) Assume reasonable value for missing data.

iii) Answer any two questions from each section in separate script.

Section – A

1. (a) Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$, to rectangular axes through the point (2, -1) and inclined at an angle $\tan^{-1}(\frac{4}{3})$. (10)

   (b) Identify the conic, $x^2 + xy + y^2 + x + y + 1 = 0$. Find also its center if it is a central conic. (10)

   (c) Find the Cartesian and spherical polar coordinates for a point whose cylindrical polar coordinates are $(5\sqrt{2}, \tan^{-1}(\frac{1}{3}), 5)$. (10)

2. (a) Determine the direction cosines of a line $3x - 2y - 8 = 0 = 6y - 3z + 15$. Also find the coordinates of the point where the line joining the points (2, - 3, 1) and (3, - 4, - 5) cuts the plane $5x + 2y + z - 8 = 0$. (17)

   (b) Test whether the four points (0, 0, 0), (3, 0, 0), (0, 4, 0) and (0, 0, - 5) are coplanar or not. If not then find the volume of a tetrahedron whose vertices are these four points. (13)

3. (a) Determine the great circle distance between Rajshahi (24°22'N, 88°35'E) and Washington (38°54'N, 77°8'W) in Kilometers. (15)

   (b) Solve the right angle triangle for which b = 135°44'17", c = 62°29'35", C = 90°. (15)

Section – B

4. (a) Define explicit function and logistic function with their graphical representations. (10)

   (b) If $A = \begin{bmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{bmatrix}$, find the matrix C satisfying the relation (10)

   A + 2C = B.

   (c) Find the following terms with examples: (10)

      (i) Idempotent Matrix

      (ii) Skew-symmetric Matrix

5. (a) Test whether the matrix $\begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}$ is unitary or not. (12)

   (b) Find the inverse of $A = \begin{bmatrix} -2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$ by using elementary row operations. (18)

6. (a) Define rank matrix. (15)

Reduce a matrix, A = $\begin{bmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{bmatrix}$ into echelon form and find its rank.

(b) What do you mean by linear equation? Solve the following system of equations with help of matrices: (15)

$$x + y + z = 6$$
$$x - y + z = 2$$
$$2x - y + 3z = 9$$

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP1st Year1stTermRegular Examination, 2020

URP1111

(History/Evolution of Human Settlement)

Full Marks: 120 Time: 1.5 hrs

N.B. i) Answer any **two** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Write down the importance of studying evolution of human settlements for the students of Urban and Regional Planning. (10)
   (b) Mention the major innovations or evolutions that occurred during the Stone Age, Bronze Age, and Iron Age. (10)
   (c) Mention the various stages in the evolution of human settlement. List out the site factors for human settlement. (10)
2. (a) What are the major types of rural settlements in Bangladesh and crucial problems faced by them? (15)
   (b) Write short notes on: (15)
   (i) Pundravarddhan,(ii) Samatata, (iii) Vanga
3. (a) Draw a "pizza slice" of the Garden City with proper labels depicting the layout. (10)
   (b) Describe the following concepts: (15)
      I. Three Magnet Theory.
      II. Vertical City
      III. Linear City
   (c) Write about the street system in Plan Voisin. (05)

Section – B

4. (a) Define setback rules. Describe the common features of town settlement in Egyptian Civilization. (15)
   (b) "Elaborate drainage and secured housing were some of the main features of city infrastructure of Indus Valley Civilization" – Justify the statement with supportive features. (15)
5. (a) How was the theory of Hippodamus implemented in ancient Greek street system? Briefly describe about Greek Acropolis. (15)
   (b) Illustrate the concept of "Arch" and "Vault" in Roman engineering technique. How "Forum" served as public space in ancient Roman Civilization? (15)
6. (a) Mention the five principles of settlement. Which factors do you consider for the description of the settlement? (15)
   (b) Why did Mesopotamian Civilization grow? Describe the locational and attributional features of the UR City of Mesopotamian Civilization (15)